

D4: Full ELBO Under $\eta = \mathbf{YF}\tilde{\beta}$

Cluster B (Cox-on-YF Reformulation) — Derivation D4 (Priority)

Identifies all changes to `code/compute_elbo.R`

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Contents

1	Context and Goal	2
2	Full ELBO Structure	2
3	Term (I): Genomics Expected Log-Likelihood	2
4	Term (II): Survival Expected Log-Likelihood (Updated)	2
4.1	Cox PH Taylor Approximation	2
4.2	Variance of η_i Under Cluster B	3
5	Term (III): KL for $q(\mathbf{L})$ (Simplified)	3
6	Term (IV): KL for $q(\mathbf{F})$ (Dual-Source)	3
7	Term (V): KL for $q(\tilde{\beta})$	4
8	Term (VI): KL for $q(\tau)$	4
9	Summary of ELBO Changes	4
10	Code Changes to <code>compute_elbo.R</code>	4
10.1	<code>compute_survival_elbo()</code> — Updated	4
10.2	<code>compute_ebnm_kl()</code> — Unchanged	5
10.3	Caller: <code>fit_modular.R</code>	5
11	Verification Checklist	5

1 Context and Goal

This document derives the full Evidence Lower BOund (ELBO) for the Cluster B model $\eta_i = (\mathbf{y}_i \mathbf{F}) \tilde{\beta}$. The ELBO is needed for convergence monitoring and to verify that the CAVI updates are improving the correct objective. The proxy used in the original code (τ -based reconstruction error + EBNM KL terms) must be updated to reflect the new parameterization.

We identify exactly which functions in `code/compute_elbo.R` need updating.

2 Full ELBO Structure

$$\mathcal{L} = \underbrace{\mathbb{E}_q[\log p(\mathbf{Y} \mid \mathbf{L}, \mathbf{F}, \boldsymbol{\tau})]}_{\text{(I) genomics}} + \underbrace{\mathbb{E}_q[\log p(\mathbf{t}, \boldsymbol{\delta} \mid \mathbf{Y}, \mathbf{F}, \tilde{\beta})]}_{\text{(II) survival}} - \underbrace{\text{KL}[q(\mathbf{L}) \parallel p(\mathbf{L})]}_{\text{(III) KL-L}} - \underbrace{\text{KL}[q(\mathbf{F}) \parallel p(\mathbf{F})]}_{\text{(IV) KL-F}} - \underbrace{\text{KL}[q(\tilde{\beta}) \parallel p(\tilde{\beta})]}_{\text{(V) KL-}\tilde{\beta}} - \underbrace{\text{KL}[q(\boldsymbol{\tau}) \parallel p(\boldsymbol{\tau})]}_{\text{(VI) K}} \quad (1)$$

Terms (I), (III), (VI) are structurally identical to the original model (same formulas, just with $q(\mathbf{L})$ now pure-genomics and $q(\boldsymbol{\tau})$ unchanged). Terms (II), (IV), (V) change under Cluster B.

3 Term (I): Genomics Expected Log-Likelihood

$$\mathbb{E}_q[\log p(\mathbf{Y} \mid \mathbf{L}, \mathbf{F}, \boldsymbol{\tau})] = -\frac{np}{2} \log(2\pi) + \frac{1}{2} \sum_j n \log \tau_j - \frac{1}{2} \sum_{i,j} \tau_j \mathbb{E}_q[(y_{ij} - \mathbf{l}_i^\top \mathbf{f}_j)^2]. \quad (2)$$

The expected squared residual is

$$\mathbb{E}_q[(y_{ij} - \mathbf{l}_i^\top \mathbf{f}_j)^2] = (y_{ij} - \mathbb{E}[\mathbf{l}_i]^\top \mathbb{E}[\mathbf{f}_j])^2 + \sum_k \left(\mathbb{E}[l_{ik}^2] \mathbb{E}[f_{jk}^2] - \mathbb{E}[l_{ik}]^2 \mathbb{E}[f_{jk}]^2 \right).$$

This is computed by `update_tau()` as the ELBO proxy; the formula is unchanged from the original model.

Term (I) status

Unchanged. The same formula applies under Cluster B; $\mathbf{L}$ is still the genomics latent variable, just no longer coupled to survival.

4 Term (II): Survival Expected Log-Likelihood (Updated)

4.1 Cox PH Taylor Approximation

The Cox partial log-likelihood $\log \text{PL}(\boldsymbol{\eta})$ is non-conjugate. We use the second-order Taylor expansion around the current estimate $\hat{\boldsymbol{\eta}}$:

$$\log \text{PL}(\boldsymbol{\eta}) \approx \log \text{PL}(\hat{\boldsymbol{\eta}}) + \mathbf{u}^\top (\boldsymbol{\eta} - \hat{\boldsymbol{\eta}}) - \frac{1}{2} (\boldsymbol{\eta} - \hat{\boldsymbol{\eta}})^\top \mathbf{W} (\boldsymbol{\eta} - \hat{\boldsymbol{\eta}}),$$

where $\mathbf{u}$ is the score vector and $\mathbf{W} = \text{diag}(w_1, \dots, w_n)$ the negative diagonal Hessian. Evaluated at the variational expectation $\mathbb{E}_q[\boldsymbol{\eta}]$:

$$\begin{aligned} \mathbb{E}_q[\log \text{PL}] &\approx \log \text{PL}(\hat{\boldsymbol{\eta}}) + \mathbf{u}^\top (\mathbb{E}_q[\boldsymbol{\eta}] - \hat{\boldsymbol{\eta}}) - \frac{1}{2} \sum_i w_i \mathbb{E}_q[\eta_i^2] + \frac{1}{2} \sum_i w_i \hat{\eta}_i^2 \\ &= \log \text{PL} - \frac{1}{2} \sum_i w_i \text{Var}_q(\eta_i) + \text{terms in } \hat{\boldsymbol{\eta}}, \end{aligned} \quad (3)$$

where $\log\text{PL} = \log\text{PL}(\hat{\boldsymbol{\eta}})$ is the partial log-likelihood at the current posterior means and is computed once per outer CAVI iteration.

4.2 Variance of η_i Under Cluster B

Under the original model, $\eta_i = \sum_k l_{ik} \beta_k$, so

$$\text{Var}_q(\eta_i) = \sum_k \left(\mathbb{E}[l_{ik}^2] \mathbb{E}[\beta_k^2] - \mathbb{E}[l_{ik}]^2 \mathbb{E}[\beta_k]^2 \right).$$

Under Cluster B, $\eta_i = \sum_k (z_F)_{ik} \tilde{\beta}_k$ where $(z_F)_{ik}$ is treated as a fixed constant per outer iteration. Therefore

$$\text{Var}_q(\eta_i) = \sum_k (z_F)_{ik}^2 \text{Var}_q(\tilde{\beta}_k) = \sum_k (z_F)_{ik}^2 \left(\mathbb{E}[\tilde{\beta}_k^2] - \mathbb{E}[\tilde{\beta}_k]^2 \right). \quad (4)$$

Key simplification

Because $\mathbf{Z}_F$ is observed (not latent), the cross terms $\text{Var}_q(l_{ik} \beta_k) = \text{Var}_q(l_{ik}) \mathbb{E}[\beta_k^2] + \text{Var}_q(\beta_k) \mathbb{E}[l_{ik}^2]$ that appear in the original formulation collapse to just $\text{Var}_q(\tilde{\beta}_k) (z_F)_{ik}^2$. No second moment of $\mathbf{L}$ is needed in the survival ELBO.

The survival ELBO term becomes:

$$\begin{aligned} \mathbb{E}_q[\log p(\mathbf{t}, \boldsymbol{\delta} \mid \mathbf{Y}, \mathbf{F}, \tilde{\boldsymbol{\beta}})] &\approx \log\text{PL} - \frac{1}{2} \sum_i w_i \sum_k (z_F)_{ik}^2 \text{Var}_q(\tilde{\beta}_k) \\ &= \log\text{PL} - \frac{1}{2} \mathbf{1}^\top \left[\mathbf{Z}_F^2 \left(\mathbf{E}[\tilde{\boldsymbol{\beta}}^2] - \mathbf{E}[\tilde{\boldsymbol{\beta}}]^2 \right) \right] \circ \mathbf{w}, \end{aligned} \quad (5)$$

where $\mathbf{Z}_F^2$ denotes element-wise squaring of $\mathbf{Z}_F$ and $\mathbf{E}[\tilde{\boldsymbol{\beta}}^2] - \mathbf{E}[\tilde{\boldsymbol{\beta}}]^2$ is the K-vector of posterior variances of $\tilde{\boldsymbol{\beta}}$.

5 Term (III): KL for $q(\mathbf{L})$ (Simplified)

$q(\mathbf{L})$ is now a pure-genomics EBNM posterior (D1). The KL is computed by `compute_ebnm_kl()` in `code/compute_elbo.R`, which takes the EBNM log-likelihood and the pseudo-likelihood components as inputs. This formula is unchanged; only the EBNM inputs (A_L and B_L) change (now genomics-only, from D1).

Term (III) status

Formula unchanged. `compute_ebnm_kl()` implementation unchanged. The KL value changes because EBNM inputs change (smaller A_L , no survival contribution).

6 Term (IV): KL for $q(\mathbf{F})$ (Dual-Source)

$q(\mathbf{f}_k)$ is now a dual-source EBNM posterior (D2). The KL is still computed by `compute_ebnm_kl()`, which is parameterization-agnostic — it takes the EBNM log-likelihood and the EBNM inputs $A_{F,j}$, $x_{F,j}$, $\mathbb{E}[f_{jk}]$, $\mathbb{E}[f_{jk}^2]$. Since those inputs now include the survival contribution, the KL value changes, but the `compute_ebnm_kl()` function itself does not need modification.

Term (IV) status

Formula unchanged. `compute_ebnm_kl()` implementation unchanged. The KL value changes because $q(\mathbf{F})$ is now dual-source (larger effective precision $A_{F,j}$, different posterior shape).

7 Term (V): KL for $q(\tilde{\beta})$

The $q(\tilde{\beta}_k)$ EBNM update uses the same scalar EBNM structure as the original $q(\beta_k)$, with the predictor changed from l_{ik} to $(z_F)_{ik}$ (D3). The `compute_ebnm_kl()` function is called with the new $A_{\tilde{\beta},k}$, $x_{\tilde{\beta},k}$, $\mathbb{E}[\tilde{\beta}_k]$, $\mathbb{E}[\tilde{\beta}_k^2]$. No change to the KL formula itself.

Term (V) status

Formula unchanged. `compute_ebnm_kl()` implementation unchanged. KL value changes because $A_{\tilde{\beta},k}$ now uses $(z_F)_{ik}^2$ instead of $\mathbb{E}[l_{ik}^2]$.

8 Term (VI): KL for $q(\tau)$

The $q(\tau)$ update is completely unchanged (D4 does not affect it). The KL contribution is embedded in the `update_tau()` return value (`elbo_proxy`). This is unchanged.

9 Summary of ELBO Changes

ELBO Term	Original	Cluster B Change
(I) Genomics	$-\frac{1}{2} \sum_{i,j} \tau_j \mathbb{E}[(y_{ij} - \mathbf{I}_i^\top \mathbf{f}_j)^2]$	Unchanged
(II) Survival	$\log\text{PL} - \frac{1}{2} \sum_i w_i \sum_k \mathbb{E}[l_{ik}^2] \mathbb{E}[\beta_k^2] - \dots$	$\log\text{PL} - \frac{1}{2} \sum_i w_i \sum_k (z_F)_{ik}^2 \text{Var}_q(\tilde{\beta}_k)$
(III) KL-L	EBNM KL with dual-source inputs	EBNM KL with genomics-only inputs (simpler)
(IV) KL-F	EBNM KL with genomics-only inputs	EBNM KL with dual-source inputs
(V) KL- $\tilde{\beta}$	EBNM KL with $\mathbb{E}[l^2]$ predictor	EBNM KL with $(z_F)^2$ predictor
(VI) KL- τ	Embedded in <code>elbo_proxy</code>	Unchanged

10 Code Changes to `compute_elbo.R`

10.1 `compute_survival_elbo()` — Updated

`compute_survival_elbo()` — new signature and body

Old signature:

`compute_survival_elbo(logPL, w, EL, EL2, EBeta, EBeta2)`

New signature:

```
compute_survival_elbo(logPL, w, ZF, EBeta, EBeta2)
```

Old body (var_eta used EL second moments):

```
var_eta <- rowSums(sweep(EL2, 2, EBeta2, "*")) -  
               rowSums(sweep(EL^2, 2, EBeta^2, "*"))  
logPL - 0.5 * sum(w * var_eta)
```

New body (var_eta uses ZF and beta posterior variances):

```
# Var_q(eta_i) = sum_k (ZF[i,k])^2 * Var_q(beta_tilde_k)  
#               = sum_k ZF[i,k]^2 * (EBeta2[k] - EBeta[k]^2)  
var_tilde_beta <- EBeta2 - EBeta^2 # K-vector: Var_q(beta_tilde_k)  
var_eta <- as.vector(ZF^2 %*% var_tilde_beta) # n-vector  
logPL - 0.5 * sum(w * var_eta)
```

10.2 compute_ebnm_kl() — Unchanged

The KL computation function `compute_ebnm_kl(ebnm_log_lik, A, x, mean_q, second_q)` is called identically for all three latent variables. No code change needed.

10.3 Caller: fit_modular.R

The call at line 526 changes from:

```
surv_elbo <- compute_survival_elbo(logPL, w, EL, EL2, EBeta, EBeta2)
```

to:

```
surv_elbo <- compute_survival_elbo(logPL, w, ZF, EBeta, EBeta2)
```

11 Verification Checklist

D4 Verification

1. $\text{Var}_q(\eta_i) = \sum_k (z_F)_{ik}^2 \text{Var}_q(\tilde{\beta}_k)$ — derived in §3.2. ✓
2. When $\text{Var}_q(\tilde{\beta}_k) = 0$ (zero posterior uncertainty): survival ELBO = logPL exactly. ✓
3. When $ZF = EL$ and $EBeta2 \approx EBeta^2$: new formula approaches old formula. ✓ (sanity)
4. `compute_survival_elbo()` signature updated; old `EL`, `EL2` removed. (to implement)
5. Full ELBO is monotone increasing across CAVI iterations after code changes. (to verify in Step 9 smoke fit)
6. ELBO matches proxy (`elbo_proxy` from `update_tau`) in early iterations when survival contribution small. (to verify)